

MOHAMMED ZAKWAN M

Bangalore, Karnataka | +91 7892282514 | zakwan1292003@gmail.com
[linkedin.com/in/mohammed-zakwan-m](https://www.linkedin.com/in/mohammed-zakwan-m) | github.com/zakwanmunawar | zakwan-portfolio-zeta.vercel.app

TECHNICAL SKILLS

Core Programming: Python (Proficient), C, Java, SQL, JavaScript, HTML/CSS (Fundamentals)

Computer Vision & AI/ML: Object Detection (YOLO Architectures v1–v26), From-Scratch Model Implementation, Image Processing, Deep Learning Pipelines, Neural Networks

Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib

Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Antigravity

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems

EDUCATION

Bachelor of Technology in Computer Engineering (Artificial Intelligence & Machine Learning)

2023 – 2027

Presidency University, Bangalore, Karnataka | CGPA: 7.24 / 10.0

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Storage Architecture

PROJECTS

Aerial & UAV Object Detection & Tracking System

Python, PyTorch, YOLO, OpenCV, TensorRT, ONNX | github.com/zakwanmunawar/AeroDrop-Tracker

- Built and trained a YOLO26n detection model (2.37M parameters) on a 12-class aerial dataset with 86,000+ annotations, reaching 22.1% mAP@50 despite heavy background clutter and dense object clusters.
- Tuned the inference pipeline on an NVIDIA Tesla T4 GPU to hit sub-3ms frame latency, pushing raw throughput past 330 FPS.
- Handled preprocessing end-to-end — frame resizing, bounding-box normalization, and augmentation — to keep accuracy stable across changing backgrounds.
- Wired up OpenCV for live video inference so detections track smoothly frame-to-frame with minimal lag.

Construction Worker Safety & PPE Compliance Detector

Python, YOLO, OpenCV | github.com/zakwanmunawar/worker-safety-system

- Fine-tuned a 3M-parameter YOLO model to flag site-safety violations — falls, missing helmets, missing jackets — hitting 97.0% mAP@50 and 96.5% precision.
- Ran the full pipeline on a Tesla T4 GPU at sub-9ms latency (117+ FPS end-to-end), fast enough for live compliance alerts.
- Added color-coded bounding-box overlays in OpenCV so missing gear gets flagged visually in the live feed, without manual review.

CERTIFICATIONS & ACHIEVEMENTS

YOLO Architecture Implementation & Research Series (v1 to v26)

PyTorch & Google Colab

- Rebuilt YOLOv1 from scratch in PyTorch, then worked forward through later versions to understand what changed and why at each step.

Virtual Internship Certification in Python Programming – CodSoft 2025

- Completed a 4-week virtual internship, writing modular Python scripts and practicing core OOP and algorithmic concepts.

Data Structures & Algorithms in Python – NeetCode 75

- Worked through the NeetCode 75 problem set in Python, covering core patterns — two pointers, sliding window, graphs, dynamic programming — across all major topics.

Top 70 Selection (out of 500+ Teams) – University Innovation & Project Review (IPR)

- Built the hardware-software integration for an autonomous robotic car arm, one of 70 teams selected out of 500+.